Targeting Scar States Using Polynomial Filtering in the PXP Hamiltonian

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Eigenstate Thermalization Hypothesis (ETH)

- Explains thermalization in isolated quantum many-body systems.
- Local observables relax to match expected values from statistical mechanics.
- Individual generic eigenstates mimic thermal ensembles:

$$\langle E_n|\hat{O}|E_n\rangle \approx \langle \hat{O}\rangle_{\text{thermal}}. \tag{1}$$

Important: ETH Violations

Some systems have rare, unusual eigenstates that strongly violate ETH and fail to thermalize.

The PXP Model & Quantum Many-Body Scars

- Models projected spin-flip dynamics on a 1D Rydberg atom chain:

$$H = \sum_{i=0}^{L-1} P_i X_{(i+1)\%L} P_{(i+2)\%L} \tag{2}$$

  where $P_i = \frac{1}{2}(\mathbb{I} - Z_i)$ projects site i onto its ground state $|0\rangle$.
- ETH-violating eigenstates form **quantum many-body scar states**.
- Scars are rare, evenly spaced, and possess unusually low entanglement.
- But there is the usual problem, the Hilbert space grows exponentially.

Problem of the Project

Can we use Chebyshev polynomial filtering to target the scar states without doing full exact diagonalization?

Understanding the PXP Hamiltonian

- The Rydberg blockade excludes configurations with adjacent excitations, such as $|\cdots 11 \cdots\rangle$.
- Basis states are represented as nodes in an adjacency graph.
- The physical blockade-constrained subspace appears as the largest connected component of this graph.

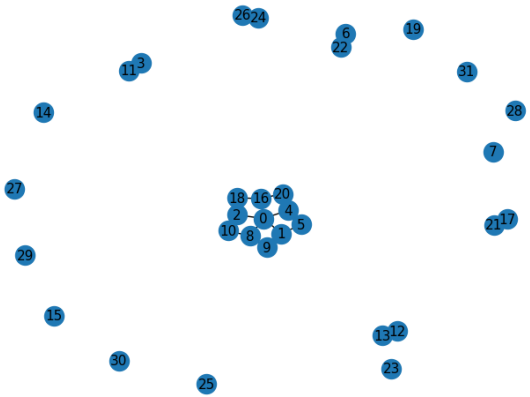


Figure: Full computational basis adjacency graph for an $L = 5$ chain.

PXP Hamiltonian is Sparse

$$H = \begin{pmatrix}
\cdot & 1 & 1 & \cdot & 1 & \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
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\end{pmatrix} \tag{3}$$

The Constrained Dimension Grows as a Fibonacci Sequence

For open boundary conditions, the blockade constraint gives the recursion

$$\mathcal{D}_{\text{open}}(L+1) = \mathcal{D}_{\text{open}}(L) + \mathcal{D}_{\text{open}}(L-1). \tag{4}$$

The two terms correspond to adding $|0\rangle$ freely, or adding $|1\rangle$ after a forced neighboring $|0\rangle$. With $\mathcal{D}_{\text{open}}(1) = 2$ and $\mathcal{D}_{\text{open}}(2) = 3$,

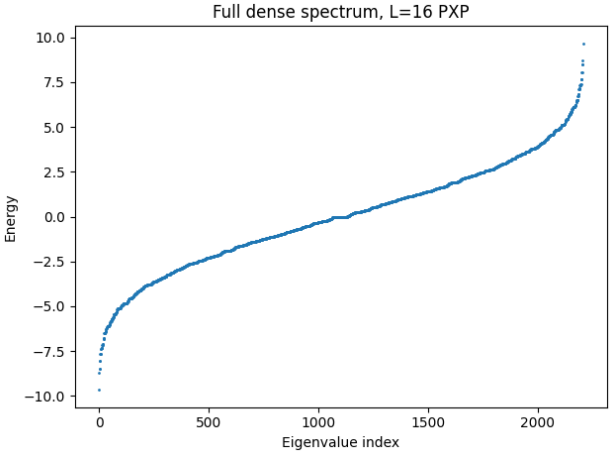
$$\mathcal{D}_{\text{open}}(L) \sim \phi^L, \qquad \phi = \frac{1+\sqrt{5}}{2}. \tag{5}$$

For periodic boundary conditions, the first and last sites are also constrained:

$$\mathcal{D}_{\text{periodic}}(L) = F_{L+1} + F_{L-1}. \tag{6}$$

Reference Benchmark via Full Exact Diagonalization

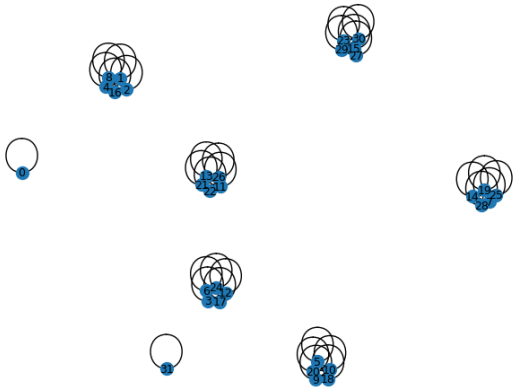
- Dense solvers (`scipy.linalg.eigh`) compute the full spectrum.
- For system size $L = 16$, the constrained sector dimension is $\mathcal{D} = 2207$.
- This exact spectrum will be our benchmark for later numerical methods.



Translation Symmetry Momentum Splitting

- Periodic chains commute with translation: $[H, T] = 0$.
- We decompose space into momentum blocks $k = 2\pi n_k / L$.
- Symmetrized basis states built from orbit families $\mathcal{F}(\vec{n})$:

$$|\psi_{k,\vec{n}}\rangle = \frac{1}{\mathcal{N}_{k,\vec{n}}} \sum_{r=1}^{L} e^{ikr}\, T^r\, |\mathrm{rep}(\vec{n})\rangle\,. \tag{7}$$



Orbit Selection Rules & Momentum Sector Diagonalization

When we look at the normalization for $f_{\vec{n}} < L$, the normalization condition gives us

$$f_{\vec{n}} n_k \pmod{L} = 0. \tag{8}$$

If this rule is not satisfied, then the projected momentum state must be omitted from the basis.
For normalizable states,

$$|\psi_{k,\vec{n}}\rangle = \frac{1}{\sqrt{f_{\vec{n}}}} \sum_{r=1}^{f_{\vec{n}}} e^{ikr} T^r \, |\mathrm{rep}(\vec{n})\rangle \, . \tag{9}$$

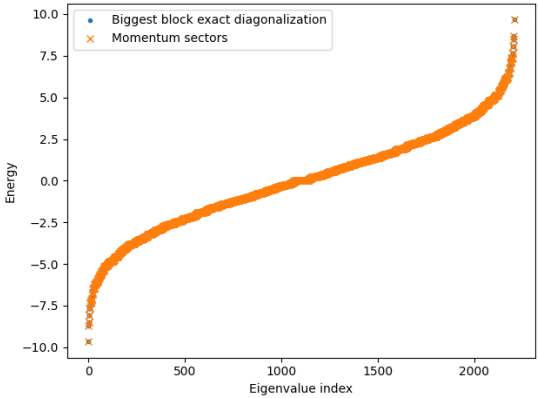


Figure: Exact energy eigenvalues resolved across isolated momentum sectors.

Entanglement Entropy via SVD

We focus on the $k = 0$ momentum block eigenstates to see the scar states. To find the singular values, we first reshape

$$V_{k=0} = \begin{pmatrix} \uparrow & \uparrow & & \uparrow \\ |\psi_1\rangle & |\psi_2\rangle & \cdots & |\psi_N\rangle \\ \downarrow & \downarrow & & \downarrow \end{pmatrix} \implies |\psi_i\rangle = \begin{pmatrix} \psi^{(i)}_{1,1} \\ \psi^{(i)}_{1,2} \\ \vdots \\ \psi^{(i)}_{2^{L_A},2^{L_B}} \end{pmatrix} \xrightarrow{\text{reshape}} M^{(i)} = \begin{pmatrix} \psi^{(i)}_{1,1} & \cdots & \psi^{(i)}_{1,2^{L_B}} \\ \vdots & \ddots & \vdots \\ \psi^{(i)}_{2^{L_A},1} & \cdots & \psi^{(i)}_{2^{L_A},2^{L_B}} \end{pmatrix}.$$

Then we find the entanglement entropies after applying SVD to all $M^{(i)}$

$$S_A = -\sum_{\alpha} s_\alpha^2 \log s_\alpha^2. \tag{10}$$

States with unusually low entropy: Scar States

- Thermal states have high entropies. Their high entanglement results in loss of local information.
- Scar states have unusually low entropies.
- Out of 143 total number of eigenstates in $k = 0$, only 8 of them are scar states.
- The scar state energies span the entire energy range!

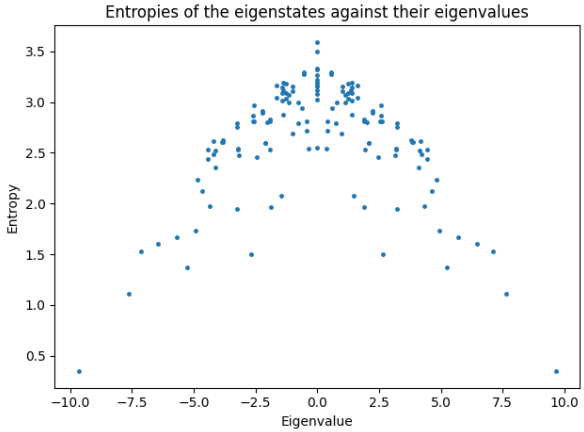


Figure: Entropies S_A vs energy for $k = 0$ at $L = 16$.

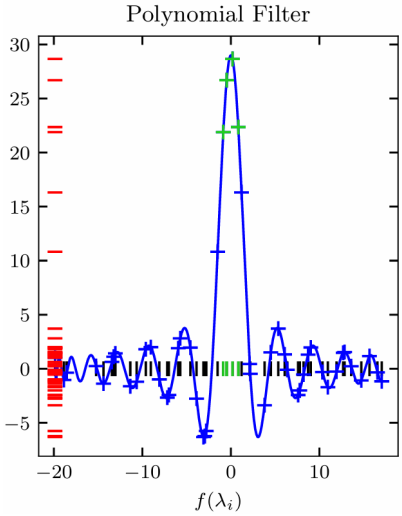
The Beauty of Chebyshev Polynomial Filtering

- There is a way to target scar states without full exact diagonalization.
- We target an energy eigenstate, amplify its eigenvalue, and suppress the rest.
- The ideal object would be

$$\delta(E - H), \tag{11}$$

- We can approximate it by

$$p_k(H) = \sum_{m=0}^{k} c_m T_m(H). \tag{12}$$



Implementation of Chebyshev Polynomial Filtering

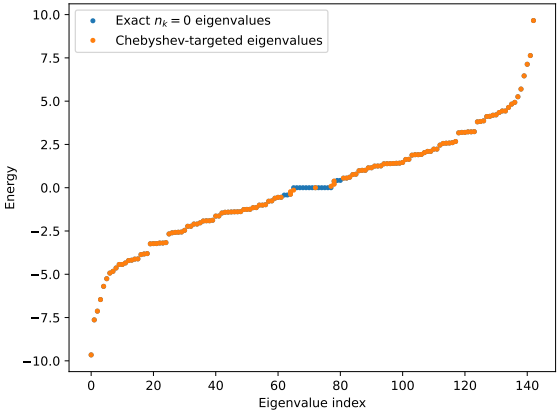
- To be able to reach larger system sizes, we keep everything sparse, and define the recursive method

$$p_k(H)\vec{x} = \sum_{m=0}^{k} c_m T_m \left(1 + \frac{H - \lambda_{\max}}{\lambda_{\max}}\right) \vec{x}$$

- We sweep over

$$E_{\text{target}} = -(L+1), \ldots, L+1.$$

- Find 20 eigenpairs around each E_{target} with a sparse eigensolver.



Limitations of the Implementation

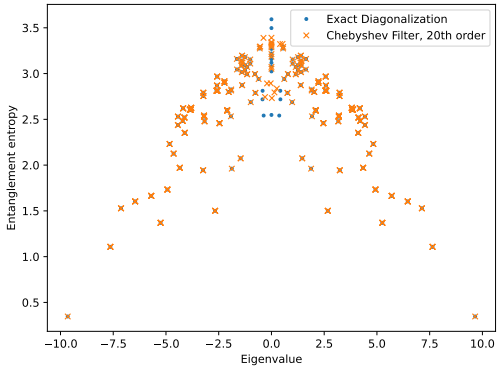


Figure: Chebyshev filtering vs exact diagonalization with order $k = 20$.

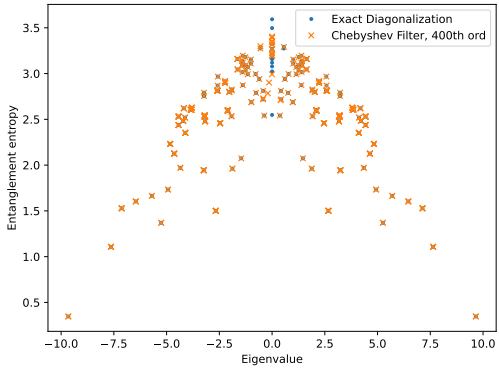
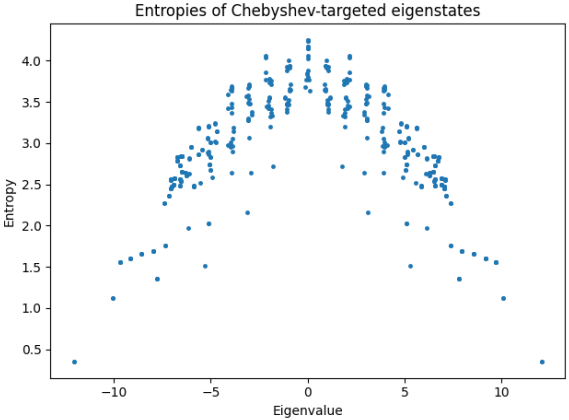


Figure: Chebyshev filtering vs exact diagonalization with order $k = 400$.

Largest system size reached with Chebyshev filtering

- Largest system size achieved was $L = 20$.
- Full ED is still reliable at $L = 20$ for $k = 0$ momentum sector.
- For even larger L, Chebyshev filtering is expected to be better.
- Real bottleneck of this implementation is the basis states initialization

$$H \to H_{\text{biggest}} \to (H_{\text{biggest}})_{k=0}.$$



Summary and Future Outlook

- The PXP Hamiltonian contains rare ETH-violating scar states with unusually low entanglement.
- Translation symmetry reduces the problem to independent momentum sectors, we focus on $k = 0$ block for finding scar states.
- Chebyshev polynomial filtering can target eigenstates near chosen energies without diagonalizing the full spectrum.
- The method reproduces the benchmark eigenspectrum and entropy profile, but the current bottleneck is still basis initialization.
- Future work: construct the blockade-constrained $k = 0$ basis directly and study time evolution of *Néel type* states.

Chebyshev polynomial filtering implementation

Following the lecture notes, we use the filter

$$p_k(H) = \sum_{m=0}^{k} c_m T_m \left(1 + \frac{H - \lambda_{\max}}{\lambda_{\max}}\right), \tag{13}$$

where the lecture note coefficients target $E_{\text{target}} = 0$:

$$c_m = \sqrt{4 - 3\delta_{m0}} \cos\left(\frac{m\pi}{2}\right). \tag{14}$$

Since $\arccos(0) = \pi/2$, we generalize this to a target energy E_{target} by replacing

$$0 \to \frac{E_{\text{target}}}{\lambda_{\max}}. \tag{15}$$

Therefore,

$$c_m = (2 - \delta_{m0}) \cos\left[m \arccos\left(\frac{E_{\text{target}}}{\lambda_{\max}}\right)\right]. \tag{16}$$